

motorBench 2.56.0 Release Notes

READY

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Overview of motorBench® Development Suite

motorBench® Development Suite is a graphical, interactive development environment designed to help motor control engineers to develop and implement motor control systems on Microchip dsPIC® Digital Signal Controllers. motorBench provides features to:

- Select between start-up methods, rotor position feedback options and additional control features supported by the Motor Control Application Framework (MCAF)
- Measure and report the electrical parameters of the motor and the mechanical parameters of the system
- Quickly obtain stable Proportional-Integral (PI) control loop gains for velocity, flux, and torque
- See how the control loop gains affect the system through Bode plots
- Generate code straight into an MPLAB X IDE project to spin the motor

What's New

Motor Control Application Framework (MCAF) R9 — see [MCAF User's Guide](#)¹ for more information.

- Added support for new dsPIC33A and dsPIC33CK devices, specifically:
 - dsPIC33AK(256/512)MPS2xx
 - dsPIC33AK(256/512)MPS5xx
 - dsPIC33AK(128/256)MPS1xx
 - dsPIC33AK(128/256)MPS3xx
 - dsPIC33CK(64/128/256)MC00x
- Added motor parameter files for two new Leadshine reference motors
 - Leadshine 200W 24V BLDC motor with 2500 PPR encoder
 - Leadshine 400W 220 V_{AC} Servo Motor with 2500 PPR quadrature encoder
- Bug fixes and improvements
 - Added the ability to disable ADC gain compensation for dsPIC33A devices
 - Added fault input pin reference to MCLV-48V-300W board config for PC1xR fault sources

System Requirements

This release of motorBench® Development Suite has been tested with the following versions of development tools.

- MPLAB X IDE v6.35
- Compiler: XC-DSC v4.00.00

DFP Family	Version
dsPIC33CK-MP_DFP	1.15.423
dsPIC33CK-MC_DFP	1.10.386
dsPIC33AK-MP_DFP	1.4.260
dsPIC33AK-MC_DFP	1.5.263
dsPIC33E-GM-GP-MC-GU-MU_DFP	1.6.297

The following versions of MCC components are recommended to be used with this version of motorBench.

MPLAB Code Configurator® (MCC) Components	Version
MCC plugin	5.7.0
MCC core	5.9.0
MCC content manager	7.0.0

¹ <https://microchiptech.github.io/mcaf-doc/latest/>

MPLAB Code Configurator® (MCC) Components	Version
MCC Melody core	2.10.1

For other compatible MCC content versions please refer to the appropriate manifest file hosted in the [motorBench release collateral repository](#)².

Supported Hardware

This release of motorBench® Development Suite supports both low-voltage and high-voltage setups.

Note

Motor parameter measurement (Self-Commissioning) is supported only on dsPIC33EP256MC506 External Op Amp Motor Control PIM and MCLV-2 Development Board.



For use with MPLAB X v6.25 and later, please upgrade dsPICDEM MCHV-2 Development Board ([DM330023-2](#)³) and dsPICDEM MCHV-3 Development Board ([DM330023-3](#)⁴) to replace the onboard isolated programmer on these two boards with the High Voltage Motor Control Isolated Debugger Card ([DV43T08A](#)⁵).

Without this upgrade, motorBench v2.56.0 will have partial support for the stock versions of these two boards:

-  Motor parameter measurement is not supported
-  Programming via ICD connection is not supported; it requires MPLAB X IDE v.6.20 or earlier
-  All other features of motorBench are supported, including generating code and using MPLAB X to compile and link the code (MPLAB Integrated Programming Environment (IPE) 6.20 may be used to program a device from the resulting .hex file.)

MCS MCHV-230VAC-1.5kW Motor Control High-Voltage Development Board ([EV78U65A](#)⁶) is the recommended board for all new development work that is AC mains-powered.



dsPIC33E family support

Support for the dsPIC33E family of devices is deprecated: although this family is supported in motorBench v2.56.0, support for dsPIC33E devices will be removed in a future version of motorBench.

Supported Processor Plug-In Modules and DIMs

- dsPIC33EP256MC506 External Op Amp Motor Control PIM [Part Number: [MA330031-2](#)⁷] with silicon revision A8

² <https://github.com/microchip-pic-avr-solutions/motorbench-release-collateral/tree/2.56.0>

³ <http://www.microchip.com/DM330023-2>

⁴ <http://www.microchip.com/DM330023-3>

⁵ <https://www.microchip.com/en-us/development-tool/dv43t08a>

⁶ <https://www.microchip.com/ev78u65a>

⁷ <https://www.microchip.com/en-us/development-tool/MA330031-2>

- dsPIC33CK256MP508 External Op Amp Motor Control PIM [Part Number: [MA330041-1](#)⁸]
- dsPIC33CK256MP508 Internal Op Amp Motor Control PIM [Part Number: [MA330041-2](#)⁹]
- dsPIC33CK64MP105 External Op Amp Motor Control PIM [Part Number: [MA330050-1](#)¹⁰]
- dsPIC33CK64MP105 Internal Op Amp Motor Control PIM [Part Number: [MA330050-2](#)¹¹]
- dsPIC33CK64MC105 External Op Amp Motor Control PIM [Part Number: [MA330051-1](#)¹²]
- dsPIC33CK64MC105 Internal Op Amp Motor Control PIM [Part Number: [MA330051-2](#)¹³]
- dsPIC33CK256MP508 Motor Control DIM [Part Number: [EV62P66A](#)¹⁴]
- dsPIC33CK256MC006 Motor Control DIM [Part Number: [EV19U29A](#)¹⁵]
- dsPIC33CK64MC105 Motor Control DIM [Part Number: [EV03J37A](#)¹⁶]
- dsPIC33AK128MC106 Motor Control DIM [Part Number: [EV68M17A](#)¹⁷]
- dsPIC33AK512MC510 Motor Control DIM [Part Number: [EV67N21A](#)¹⁸]
- dsPIC33AK256MPS306 Motor Control DIM [Part Number: [EV22K90A](#)¹⁹]

Low-voltage hardware

1. One of the following development boards
 - a. dsPICDEM MCLV-2 Development Board [Part Number: [DM330021-2](#)²⁰]
 - i. Any of the Plug-In Modules supported by motorBench® Development Suite.
 - b. dsPIC33CK Low Voltage Motor Control Board [Part Number: [DM330031](#)²¹]
 - c. MCS MCLV-48V-300W Development Board [Part Number: [EV18H47A](#)²²]
 - i. Any of the DIMs supported by motorBench® Development Suite.
2. A three-phase PMSM or BLDC motor that is compatible with 24V, such as the Hurst 24V BLDC motor DMA0204024B101 [Part Number: [AC300022](#)²³].
3. 24V power supply [Part Number: [AC002013](#)²⁴] - ensure that this connects to AC mains using a 2-prong cable. If you have an AC002013 with a 3-prong cable, please contact Microchip.

High-voltage hardware

1. MCHV-230VAC-1.5kW Motor Control High-Voltage Development Board [Part Number: [EV78U65A](#)²⁵]
 - a. AC mains voltages of 120VAC 60Hz and 220VAC 50Hz have been tested.
 - b. Any of the DIMs supported by motorBench® Development Suite.

⁸ <https://www.microchip.com/en-us/development-tool/MA330041-1>

⁹ <https://www.microchip.com/en-us/development-tool/MA330041-2>

¹⁰ <https://www.microchip.com/en-us/development-tool/MA330050-1>

¹¹ <https://www.microchip.com/en-us/development-tool/MA330050-2>

¹² <https://www.microchip.com/en-us/development-tool/MA330051-1>

¹³ <https://www.microchip.com/en-us/development-tool/MA330051-2>

¹⁴ <https://www.microchip.com/en-us/development-tool/EV62P66A>

¹⁵ <https://www.microchip.com/en-us/development-tool/EV19U29A>

¹⁶ <https://www.microchip.com/en-us/development-tool/EV03J37A>

¹⁷ <https://www.microchip.com/en-us/development-tool/EV68M17A>

¹⁸ <https://www.microchip.com/en-us/development-tool/EV67N21A>

¹⁹ <https://www.microchip.com/en-us/development-tool/EV22K90A>

²⁰ <https://www.microchip.com/en-us/development-tool/DM330021-2>

²¹ <https://www.microchip.com/en-us/development-tool/DM330031>

²² <https://www.microchip.com/en-us/development-tool/EV18H47A>

²³ <http://www.microchip.com/AC300022>

²⁴ <https://www.microchip.com/AC002013>

²⁵ <https://www.microchip.com/ev78u65a>

2. A three-phase PMSM or BLDC motor that is compatible with rectified AC mains voltage, such as the Leadshine 400W BLDC motor EL5-M0400-1-24 [Part Number: [AC300025](#)²⁶].

Development board selection guide

There are certain reasons to prefer specific low-voltage development boards for given tasks. In brief:

- Prefer MCLV-2 for low-current motors like the [AC300020](#)²⁷ which utilize only a small fraction of the development board's output current range — otherwise the motor and drive are mismatched.
- Otherwise, prefer one of the other boards. Certain algorithms such as ZS/MT have improved performance when using the capabilities of these boards, namely sensing all three phase currents and using shorter dead times.

Board	Output current*		Module	Other features / comments
	Range	Channels		
MCLV-2 ²⁸	2.29 A	A, B, DC	100-pin PIM	Layout issue with current sense traces impacts current sense gains (see MCAF User's Guide)
dsPIC33CK LVMC ²⁹	14.1 A	A, B, C, DC	—	Temperature sensor, sub-microsecond dead time. A fixed dsPIC33CK256MP508 ³⁰ device is soldered directly to the board.
MCLV-48V-300W ³¹	16.8 A	A, B, C, DC	120-pin DIM	Temperature sensor, sub-microsecond dead time

*Current is maximum continuous motor phase current when used with motorBench® Development Suite and MCAF.

Other required hardware

For MCLV-2 Development Board:

1. A USB-to-logic-level-UART converter from the following list:
 - a. Saelig [USB-COM-U](#)³² or [USB-COM-U13](#)³³
 - b. [TRENDnet TU-S9](#)³⁴ v2.0
2. Programming tool - ICD4, PICkit4
3. Board calibration load resistors - this is optional, please see motorBench® Development Suite User's Guide for more details

²⁶ <http://www.microchip.com/AC300025>

²⁷ <https://www.microchip.com/en-us/development-tool/AC300020>

²⁸ <https://www.microchip.com/en-us/development-tool/dm330021-2>

²⁹ <https://www.microchip.com/en-us/development-tool/DM330031>

³⁰ <http://www.microchip.com/wwwproducts/en/dsPIC33CK256MP508>

³¹ <https://www.microchip.com/en-us/development-tool/EV18H47A>

³² <http://www.saelig.com/product/USSP004.htm>

³³ <http://www.saelig.com/USSP/USSP005.htm>

³⁴ <https://www.trendnet.com/products/USB-adapters/TU-S9>

Installing motorBench[®] Development Suite v2.56.0

For installation instructions please see the motorBench[®] User's Guide on the [motorBench release repository](#)³⁵.

Changes in MCAF

MCAF has been updated to R9 for motorBench 2.56, including

- Changes in R2:
 - Support for DC link compensation
 - Support for overmodulation
 - Support for wider range of low-voltage motors
 - Updated HAL for future MCHV2 support
 - Updated Motor Control Library
 - Numerous minor fixes
- Changes in R3:
 - MCC system module compatibility
 - MCHV-2 and MCHV-3 support
 - Inverter maximum current now has a 1:1 ratio with the maximum commanded dq-frame current of the drive, operating in FOC (in R2 this incorporated a derating factor)
 - Other minor fixes
- Changes in R4:
 - MCC peripheral support
 - Parameter customization
 - Quadrature encoder support
 - Added new startup method (Weathervane startup)
 - Other minor fixes
- Changes in R5:
 - Added device support for dsPIC33CK256MP508
 - Added Angle-tracking PLL (ATPLL) sensorless estimator
 - Improved motorBench Customize page support
 - Other minor fixes
- Changes in R6:
 - Added device support for dsPIC33CK64MC105, dsPIC33CK64MP105
 - Added flux control module with flux weakening (FW) and maximum torque per ampere (MTPA)
 - Added Motion Control API (MCAPI)
 - Added dead-time compensation
 - Other minor fixes
- Changes in R7:
 - Added hardware abstraction layer support for MCC Melody
 - Added support for single channel current measurement
 - Added support for Zero-speed Maximum Torque (ZS/MT) sensorless estimator for motors with rotor saliency
 - Added support for outer loop voltage control
 - Added closed-loop stopping methods
 - Added support for dsPIC33CK Low Voltage Motor Control Board
 - Added support for MCLV-48V-300W Inverter Board
 - Added support for internal op amps
 - Other minor fixes
- Changes in R8:
 - Added support for custom boards

³⁵ <https://github.com/microchip-pic-avr-solutions/motorbench-release-collateral/tree/2.56.0>

- Added support for all dsPIC33CKxxxMP/MCxxx devices
- Added support for dsPIC33CDV, dsPIC33CDVC and dsPIC33CDVL SiP devices
- Added support for MCP8021, MCP8022 and MCP8027 three-phase gate driver devices
- Added Customize page option for disabling X2Cscope real-time debugging
- Bug fixes and improvements
- Changes in R9:
 - Added support for dsPIC33A family of devices, specifically – dsPIC33AK(32/64/128)MC10x and dsPIC33AK(256/512)MC5xx devices
 - Implemented a new ADC gain compensation algorithm specifically for dsPIC33A devices
 - Added support for Sliding Mode Observer (SMO)
 - Added support for MCS MCHV-230VAC-1.5kW Development Board ([EV78U65A](#)³⁶)
 - Made progress toward MISRA compliance by fixing these rule violations:
 - Rule 1.5: Obsolescent language features shall not be used — use void in function parameter list for no-argument functions
 - Rule 15.7: All if - else if constructs shall be terminated with an else statement
 - Rule 21.1: define and undef directives shall not be used on a reserved identifier or reserved macro name
 - Added a new fault case for when the input current offset is above a specified limit and added a Customize page option to allow users to adjust this limit
 - Added Customize page options for ATPLL estimator parameters, test harness and test profiling features
 - Set the software over-current based stall detector to be turned off by default
 - Added support for remappable input pins as PWM fault source
 - Added support for board power up delay as a board property
 - Added visual indicators for "yellow light" features in Customize page
 - Various usability improvements for the ZS/MT estimator
 - Bug fixes and improvements
- Changes in R9 motorBench 2.56
 - Added support for new dsPIC33A and dsPIC33CK devices, specifically:
 - dsPIC33AK(256/512)MPS2xx
 - dsPIC33AK(256/512)MPS5xx
 - dsPIC33AK(128/256)MPS1xx
 - dsPIC33AK(128/256)MPS3xx
 - dsPIC33CK(64/128/256)MC00x
 - Added motor parameter files for two new Leadshine reference motors
 - [Leadshine 200W 24V BLDC motor with 2500 PPR encoder](#)
 - [Leadshine 400W 220 V_{AC} Servo Motor with 2500 PPR quadrature encoder](#)
 - Bug fixes and improvements
 - Added the ability to disable ADC gain compensation for dsPIC33A devices
 - Added fault input pin reference to MCLV-48V-300W board config for PCIXR fault sources

Fixed Issues

motorBench Issues

Issue Key	Summary
MBPLAN-2129	Hyperlinks on the sidebar of motorBench are fixed

³⁶ <https://www.microchip.com/en-us/development-tool/ev78u65a>

Issue Key	Summary
MBPLAN-2166	Included .xml for the new LV motor Leadshine-ELVM-200-24
MBPLAN-2167	Included .xml for the new HV motor Leadshine-ELM1S-400-220

Motor Control Issues

Issue Key	Summary
DB_MC-6338	Allow user to disable ADC gain compensation for 33A parts
DB_MC-6628	Update motorBench compatibility to include dsPIC33AK MPS family devices
DB_MC-6639	Add fault input pin reference to MCLV-48V-300W board config for PCIR fault sources

Known Issues

motorBench Issues

Issue Key	Summary	Workaround
MBPLAN-2127	Multiple clicks in quick succession before the Import/Export board dialog opens causes dialog to reopen for each received click	-
MBPLAN-2036	Compare boards diffViewer path should be saved as a per-user property and not within the MCC project	-
MBPLAN-1847	On unloading motorBench, loaded peripheral drivers (ADC, UART) are not getting unloaded	-
MBPLAN-1342	Entries in the mc3 file are not protected against collision: motorBench does not qualify its ownership	-
MBPLAN-935	Self-Commissioning does not keep serial communication link active when waiting for user response to a fault message	Restart the parameter measurement
MBPLAN-888	Self-Commissioning: calibrate button remains enabled while calibration is running	-

Issue Key	Summary	Workaround
MBPLAN-887	Self-Commissioning: NPE if COM port is locked by another application	-
MBPLAN-876	Number of pole pairs is a positive integer; the Define page for motor allows any non-negative floating-point	-
MBPLAN-673	Serial port does not get closed programmatically when MCC exits during self-commissioning	-

Motor Control Issues

Issue Key	Summary	Workaround
DB_MC-560	Speed controller exhibits chattering behavior at voltage saturation hysteresis boundary (MCAF)	-
DB_MC-1521	Velocity loop step response test exhibits more than expected overshoot with MCHV2 and Leadshine400 motor	-
DB_MC-1922	LED patterns not displayed when in the TEST_DISABLE or TEST_ENABLE states	-
DB_MC-2323	Active damping is incorrectly enabled during Weathervane startup transition state	-
DB_MC-2742	ATPLL not robust to speed commands near zero; runs away to full negative speed	-
DB_MC-2794	Default current loop bandwidth not high enough to reject disturbances caused by backemf harmonics and dead-time distortion	-
DB_MC-2873	Current loop autotune input-output delay incorrectly underestimated as 1 sample instead of 1.5 or 2 samples	-
DB_MC-3127	ADC current offset calibration process leaves behind a small amount of residual offset error	-
DB_MC-3414	Potential for integer overflow in stall_detect.c → MCAF_LowSpeedDetectInit()	-

Issue Key	Summary	Workaround
DB_MC-3425	AN1292 PLL may not lock reliably under rapid acceleration when used as a secondary estimator	-
DB_MC-3432	ZS/MT can lock onto the wrong angle during flying start	-
DB_MC-3559	Parameter conversion to fixed-point integer does not warn if gains are zero	-
DB_MC-3734	Possible bug: incompatibility between ZSMT IPC startup method and QEI synchronization	-
DB_MC-4363	Operation of the AN1292 PLL estimator and the ATPLL estimator may not be robust for certain motors at the default values of minimum and startup speed.	Increase the values of minimum and startup speed in the Customize page until motor operation is reliable.
DB_MC-4423	Customize UI groups may overlap on each other in certain cases	-
DB_MC-4774	Velocity controller step response with SMO estimator does not match simulation	-
DB_MC-5117	AN1292 PLL estimator may induce instability while transitioning into/out of flux weakening	Use another sensorless estimator (E.g., ATPLL or SMO)
DB_MC-5202	Code generation OverflowError for kfilterPiOmega variable while using ATPLL estimator and PWM frequency of 10kHz or less	Increase ATPLL estimator parameter τ_1 so $(t_{PWM}/\tau_1) < 1.0$ t_{PWM} is the PWM time period
DB_MC-5549	MCAF_MainInit() has an unbounded while loop dependent on conditions outside the microcontroller	-
DB_MC-5640	MCLV-2 board starts with a hardware over-current fault at every power up event	-
DB_MC-5962	Standard inputs direction flag is being derived from pre-rate limited and pre-test harness velocity command	-
DB_MC-6114	Linker warning for default MPLAB X settings: --fill-upper option ignored on 33AK128MC106	-

Issue Key	Summary	Workaround
DB_MC-6210	SMO not supported on dsPIC33AK: atan2 not implemented yet	-
DB_MC-6260	Application/tick timer interrupt starts executing before MCAF finishes initialization	-
DB_MC-6345	Dynamic current limit - Idrivehorizon uses Idrivemax rather than Imax, inconsistent with Customize UI	-

Experimental features

Please review the notes below for certain features in the Customize page that are marked with a yellow indicator



Feature	Notes
Current measurement: Single channel	The reconstructed phase current data is sensitive to offset errors in DC link current measurement.
Sliding Mode Observer (SMO)	This is a new feature added in MCAF R9. Velocity controller tuning may be suboptimal in certain cases. The implementation of this estimator in MCAF R9 is observed to have limited dynamic performance, especially during rapid acceleration/deceleration. The SMO is not yet supported on dsPIC33AK devices; an arctangent implementation has not been provided yet.
Flux weakening: Equation-based	Using flux weakening with AN1292 PLL estimator may induce instability while transitioning into/out of flux weakening mode.
Dead-time compensation: Per-phase	Limited support and guidance available for tuning/application of this feature.
Outer loop controller: voltage	Limited support and guidance available for tuning/application of this feature.
Dynamic current limit: simple	Limited support and guidance available for tuning/application of this feature.

Feature	Notes
Stopping mode: Closed-loop current	Limited support and guidance available for tuning/application of this feature.
Stopping mode: Closed-loop velocity	Limited support and guidance available for tuning/application of this feature.

Limitations

Software Limitations

motorBench® Development Suite is tested for serial communication using Windows 7, Windows 10 and Windows 11 platforms. Other platforms may work with standard baud rates, but this operation has not been verified.

Backward compatibility

Projects created with earlier versions of motorBench® Development Suite *may be* compatible with version 2.56.0, however this is not fully tested. After migrating an existing project to use motorBench v2.56.0, please verify the motorBench configuration - especially parameters in the Customize page.

Motor Control Limitations

Listed below are the known limitations for this release of motorBench® Development Suite:

1. **One mechanical load** - constant load. This represents a mechanical load with constant inertia, viscous damping, and friction. The velocity control loop can generally reject external disturbance torques, within the rated current of the motor and board, and within the bandwidth of the velocity control loop. Mechanical loads with time-varying or angle-varying inertia, viscous damping, and friction, such as a blower, compressor, or pump, are currently not supported.
2. **One motor type** - PMSM,
MCLV-2 (not updated for LVMC/MCLV-48V-300W):
The reference motor is the Nidec Hurst motor DMA0204024B101 (MicrochipDirect part number [AC300022](https://www.microchip.com/DevelopmentTools/ProductDetails/ac300022)³⁷). Microchip has also validated motorBench® Development Suite (including motor parameter measurement) with motors with parameters plotted below. Please also read the following section on Supported Motor Parameters. If motorBench® Development Suite is unable to spin a motor successfully, please contact Microchip staff for additional assistance.

³⁷ <https://www.microchip.com/DevelopmentTools/ProductDetails/ac300022>

	units	min	max	plot
Resistance (R)	ohms (line-line)	0.1714	5.664	
Inductance, q-axis (Lq)	mH (line-line)	0.02982	5.269	
Inductance, d-axis (Ld)	mH (line-line)	0.03576	5.172	
Back-emf (Ke)	Vrms/KRPM (line-line)	0.8808	6.49	
Friction (Tf)	mNm	0.556	54.03	
Viscous damping (B)	uNm/(rad/s)	1.012	196.9	
Inertia (J)	uNm/(rad/s ²)	0.362	1162	
Electrical time constant L/R	ms	0.1913	3.889	
Mechanical time constant 2/3 JR/Ke ²	ms	2.27	47.25	

(Note: Mechanical time constant $(2/3) \times JR/Ke^2$ represents the time constant of velocity acceleration under an open-loop synchronous-frame voltage step, neglecting the effects of inductance, with J, R, and Ke expressed in canonical metric units. R is expressed as line-neutral resistance = half of line-line resistance, and Ke is expressed as V/(rad/s) line-neutral zero-peak = $V_{rms}/KRPM$ (line-line) $\times 0.007796968$)

Note

Motor parameter limits shown above are for guidance only, they are **not** absolute limits. Select an appropriate board, enter the parameters of your motor in motorBench and then generate code to know for sure if your motor is compatible with motorBench while using the selected board.

3. **Board** - see "Scope of hardware support" section of MCAF user's guide chapter for more details.
4. **Motors should be well-matched to the board and operating voltage.** The nominal DC link voltage of the MCLV-2 board is 24V. This voltage can be changed by cutting jumper J6 and using an appropriate power supply connected to the appropriate terminals of J7. Use of a mismatched motor (for example, a 12V motor used with a 24V DC link voltage) may cause a hardware over-current fault; in this case motor parameter measurement may fail with the message "Fault Code #10: Undefined Fault". Retry with an appropriate DC link voltage.
5. **Current measurement** - The passive components for the DC link current op amp circuit are not populated by default on the dsPIC33CK256MP508 Internal Op Amp PIM (MA330041-2). Algorithms that require sampling of the DC link current, such as Single-channel current measurement, are available through the on board op amp instead.
6. **One algorithm** - FOC
7. **Estimators** - PLL, QEI, ATPLL, ZS/MT, SMO
8. **Motor parameter measurement:**
 - a. **Performance criteria adjustment is not presently supported.** This includes adjustment of phase margin and PI phase lag at crossover in the current loop; Microchip has not completed validation and documentation of these adjustments.
9. **Autotuning:**
 - a. **Performance criteria adjustment of the current loop is documented in the MCAF User's Guide. Validation of any changes from default values is required, however.** Default values are generally adequate for most motors. For more information, see "Current Loop Tuning" in the MCAF User's Guide.
 - b. **Use of performance criteria adjustment of the velocity loop is not fully documented or tested.** We recommend not adjusting phase margin or PI phase lag unless necessary; cases where this is likely to occur are large inertias where $\alpha_J = JR/LK_m^{-2} > 10$, for which an increase of phase margin is appropriate. Phase margin values between 70 and 85 degrees are recommended in this case, with larger values providing additional stability at the cost of lower velocity bandwidth.
10. **Axis management not currently implemented** - supports only one axis.
11. **Code generation:**
 - a. **PWM switching frequency is variable, but current control loop must be at the same rate** (the value entered under Board parameters PWM switching frequency must be the reciprocal of the current loop sampling rate)

- b. **Integration with external user-supplied code may involve substantial changes.** Some guidelines for this are given in the documentation for the Motor Control Application Framework. While it is possible to integrate the code generated from motorBench® Development Suite with external code, it is the responsibility of the end user to validate this combination.
- 12. **Required compiler settings:**
 - a. **Optimization**
 - `-O1` or greater; `-O0` and `-Os` will both compile without errors but do not execute fast enough to complete within the 50 microsecond ADC ISR. Note: at higher optimization levels, in-circuit debugging using MPLAB X will behave unreliably with respect to breakpoints and single-stepping through C code.
 - The "Omit frame pointer" and "Unroll loops" settings must be enabled. The "Unroll loops" option is typically unselected by default.
 - b. **Memory model:**
 - Large data model (handles using pointers, not direct addressing, to allow for more than 8K of program variables)
 - Small scalar model
 - c. **Additional options:** `-Wno-volatile-register-var -finline`
- 13. **Recommended compiler settings:**
 - a. **Additional options:** `-Wundef` See [MPLAB XC16 C Compiler User's Guide](#)³⁸ (or [MPLAB XC-DSC C Compiler User's Guide](#)³⁹) for more information.

Supported Motor Parameters

Since version 2.15, motorBench® Development Suite supports a wide range of motors, subject to the following notes:

- **Motor parameter measurement does not need to complete successfully but valid motor parameters are required.** Some motors may have too low of an inductance or resistance, and may fail motor parameter measurement.
- **Other particular issues that may cause incompatibility with motorBench® Development Suite include**
 - **Large inertia values** – in this case, increasing voltage loop phase margin may prevent stability problems. (See "Autotuning" in the Limitations section of this document.)
 - **Large back-EMF harmonics** – a quasi-sinusoidal back-emf is assumed
 - **Issues involving individual motor control algorithms**, such as PLL estimator, motor startup, or stall detection
 - **High cogging torque**
 - **Mismatch between motor and drive** (namely using a motor with current and/or voltage requirements significantly different from that of the hardware)
- **Microchip cannot guarantee that motorBench® Development Suite will work correctly with all motors.** If a particular motor does not work properly, please contact the Microchip technical support for further guidance.

³⁸ <https://ww1.microchip.com/downloads/aemDocuments/documents/DEV/ProductDocuments/UserGuides/50002071.pdf>

³⁹ <https://ww1.microchip.com/downloads/aemDocuments/documents/DEV/ProductDocuments/UserGuides/MPLAB-XC-DSC-C-Compiler-User-Guide-DS50003589.pdf>

Supplemental Notes

Board support

1. In certain cases, motorBench and MCC Melody may be unable to configure ADC and PWM peripherals to meet the exact ADC sampling time and PWM dead time specifications that are requested by the user. In such cases, MCC Melody will configure these peripherals to the closest value of the configuration that is realizable. For example, MCLV-48V-300W board needs a minimum ADC sampling time of 300ns. While loading this board for a dsPIC33AK128MC106 device based project, motorBench will work with MCC Melody to configure the ADC to the maximum realizable ADC sampling time of 14.5 TAD = 181.25ns, assuming a TAD clock of 80MHz.
2. While using dsPIC33CK64MC105 Motor Control DIM with MCLV-48V-300W board, MCAF will flag an `MCAPI_FAULT_FLAG_OVERCURRENT` fault if one/both of these two conditions come true –
 - a. Hardware over-current event occurs
 - b. DC link voltage exceeds 60V-62V threshold (as per DS50003297A-page 53).

Deprecated features

Pullout method of QEI synchronization: This was deprecated in motorBench v2.45.0 / MCAF R7 and is expected to be removed in an upcoming release of motorBench. This feature is supported only on dsPIC33C and dsPIC33E family of devices in motorBench v2.56.0.

X2Cscope support with dsPIC33A devices

For X2Cscope to operate correctly with dsPIC33A devices, please select **Pre-processed (Legacy)** option in MPLAB X IDE → Tools → Options → Embedded → Generic Settings → ELF debug session symbol load methodology (MIPS/ARM). After making this selection, rebuild the project and program to hardware before attempting to connect with X2Cscope.

Compiler warnings

Certain warnings may be reported in the MPLAB X IDE Output terminal when building a motorBench® project. These are only warnings, and do not affect code functionality.

Optimization may eliminate reads and/or writes to register variables

See [GCC 4.5.1](https://gcc.gnu.org/onlinedocs/gcc-4.5.1/gcc.pdf)⁴⁰ for more information.

To suppress this warning, the following compiler flag should be added to the **Project Properties**:

```
-Wno-volatile-register-var
```

To add this compiler flag to the project, go to **Project Properties** → **Conf:[default]** → **XC16 or XC-DSC** → **xc16-gcc or xc-dsc-gcc** and add this compiler flag to **Additional options**.

⁴⁰ <https://gcc.gnu.org/onlinedocs/gcc-4.5.1/gcc.pdf>

Development board

dsPIC33CK LVMC Development Board

The maximum DC link voltage limit for this board (26V) was chosen to stay near the rating of the connector J1. Users may choose to increase this maximum voltage limit to support cases where they provide their own power supply via terminal block J2 and cut the "net tie" to decouple the control power supply from the inverter power supply. See User Guide document DS50002927A-page 15 for more information.

Board

dsPIC33CK LVMC

PIM/DIM Not Applicable

Import Board

Export Board

Compare Boards

To create a custom board, export an existing Microchip board, edit it and then import the edited board.

Characteristics

Current

Microcontroller part number

dsPIC33CK256MP508

PWM

Voltage

Dc Link

Default	24.0	V	
Fullscale	71.3	V	
Maximum	26.0	V	
Minimum	14.0	V	
Time constant	0.0003147	s	

Phase

Identification

Peripherals

Customer Support

Users of Microchip products can receive assistance through several channels:

- Distributor or Representative
- Local Sales Office
- Field Application Engineering (FAE)
- Technical Support

Customers should contact their distributor, representative or field application engineer (FAE) for support. Local sales offices are also available to help customers. A listing of sales offices and locations is available on our web site.

Technical support is available through the web site at: <http://support.microchip.com>⁴¹

⁴¹ <http://support.microchip.com/>